Assignment: Arrays & Lists (1D/2D/3D)

List 1: Create a Python list of the first 10 natural numbers. Print the 5th element using indexing.

```
lst = list(range(1,11))
print('List:', lst)
print('5th element (index 4):', lst[4])
```

Output:

```
List: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
5th element (index 4): 5
```

List 2: Make a list of 10 random integers. Slice the list to print only the first 5 elements.

```
import random
lst = [random.randint(1,100) for _ in range(10)]
print('Full list:', lst)
print('First 5:', lst[:5])
```

Output:

```
Full list: [29, 35, 63, 22, 21, 84, 37, 100, 66, 67]
First 5: [29, 35, 63, 22, 21]
```

List 3: Create a list [1, 2, 3, 4, 5]. Append the number 6 and print the list.

```
lst = [1,2,3,4,5]
lst.append(6)
print(lst)
```

Output:

```
[1, 2, 3, 4, 5, 6]
```

List 4: Create a list [10, 20, 30, 40, 50]. Remove the element at index 2.

```
lst = [10,20,30,40,50]
print('Before:', lst)
del lst[2]
print('After:', lst)
```

Output:

```
Before: [10, 20, 30, 40, 50]
After: [10, 20, 40, 50]
```

List 5: Create a list [1,2,3,4,5]. Replace the 3rd element with 99.

```
lst = [1,2,3,4,5]
print('Before:', lst)
lst[2] = 99
print('After:', lst)
```

Output:

```
Before: [1, 2, 3, 4, 5]
After: [1, 2, 99, 4, 5]
```

List 6: Create two lists [1,2,3] and [4,5,6]. Concatenate them.

```
a=[1,2,3]
b=[4,5,6]
print('Concat:', a + b)
```

Output:

```
Concat: [1, 2, 3, 4, 5, 6]
```

List 7: Create a nested list [[1,2,3], [4,5,6], [7,8,9]]. Print the element 5.

```
nl = [[1,2,3],[4,5,6],[7,8,9]]
print('Nested:', nl)
print('Element 5:', nl[1][1])
```

Output:

```
Nested: [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
Element 5: 5
```

List 8: Create a list of numbers from 1 to 20. Slice and print only the even numbers.

```
lst = list(range(1,21))
evens = lst[1::2]
print('Evens:', evens)
```

Output:

```
Evens: [2, 4, 6, 8, 10, 12, 14, 16, 18, 20]
```

List 9: Create a list of 5 strings. Access the second and fourth elements.

```
names = ['alpha','beta','gamma','delta','epsilon']
print('2nd:', names[1])
print('4th:', names[3])
```

Output:

```
2nd: beta
4th: delta
```

List 10: Create a list of 10 numbers. Reverse the list using slicing.

```
lst = list(range(1,11))
print('Original:', lst)
print('Reversed:', lst[::-1])
```

Output:

```
Original: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
Reversed: [10, 9, 8, 7, 6, 5, 4, 3, 2, 1]
```

Array 1: Create an integer array [10,20,30,40,50]. Print the third element.

```
import array as arr
a = arr.array('i',[10,20,30,40,50])
print('Array:', a.tolist())
print('3rd element:', a[2])
```

Output:

```
Array: [10, 20, 30, 40, 50]
3rd element: 30
```

Array 2: Create an array of type 'i' (integers). Insert numbers from 1 to 5. Slice and print elements at index 1–3.

```
import array as arr
a = arr.array('i')
for i in range(1,6): a.append(i)
```

```
print('Array:', a.tolist())
print('Slice 1-3:', a[1:4].tolist())
```

Output:

```
Array: [1, 2, 3, 4, 5]
Slice 1-3: [2, 3, 4]
```

Array 3: Create an integer array [2,4,6,8,10]. Append 12 to the array.

```
import array as arr
a = arr.array('i',[2,4,6,8,10])
a.append(12)
print(a.tolist())
```

Output:

```
[2, 4, 6, 8, 10, 12]
```

Array 4: Create an array [5,10,15,20,25]. Remove the element 15.

```
import array as arr
a = arr.array('i',[5,10,15,20,25])
print('Before:', a.tolist())
a.remove(15)
print('After:', a.tolist())
```

Output:

```
Before: [5, 10, 15, 20, 25]
After: [5, 10, 20, 25]
```

Array 5: Create two arrays [1,2,3] and [4,5,6]. Extend the first array with the second

```
import array as arr
a1 = arr.array('i',[1,2,3])
a2 = arr.array('i',[4,5,6])
a1.extend(a2)
print(a1.tolist())
```

Output:

```
[1, 2, 3, 4, 5, 6]
```

Array 6: Create an array [1,2,3,4,5]. Update the 2nd element to 99.

```
import array as arr
a = arr.array('i',[1,2,3,4,5])
print('Before:', a.tolist())
a[1]=99
print('After:', a.tolist())
```

Output:

```
Before: [1, 2, 3, 4, 5]
After: [1, 99, 3, 4, 5]
```

Array 7: Create an array [10,20,30,40,50]. Use slicing to print the first three elements.

```
import array as arr
a = arr.array('i',[10,20,30,40,50])
print(a[0:3].tolist())
```

Output:

```
[10, 20, 30]
```

Array 8: Create an array [100,200,300,400]. Reverse the array using slicing.

```
import array as arr
a = arr.array('i',[100,200,300,400])
print('Reversed:', a[::-1].tolist())
```

Output:

```
Reversed: [400, 300, 200, 100]
```

Array 9: Create an array [5,10,15,20,25,30]. Print every second element using slicing.

```
import array as arr
a = arr.array('i',[5,10,15,20,25,30])
print(a[::2].tolist())
```

Output:

```
[5, 15, 25]
```

Array 10: Create an array [2,4,6,8,10]. Convert it to a Python list and print the list.

```
import array as arr
a = arr.array('i',[2,4,6,8,10])
print(list(a))
```

Output:

```
[2, 4, 6, 8, 10]
```

Numpy 1D-1: Create a 1D array of numbers from 0 to 20. Print the 5th element.

```
import numpy as np
a = np.arange(0,21)
print('Array:', a)
print('5th element (index 4):', a[4])
```

Output:

```
Array: [ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]
5th element (index 4): 4
```

Numpy 1D-2: Create a 1D array of the first 15 odd numbers. Slice elements from index 3 to 8.

```
import numpy as np
odds = np.arange(1,30,2)[:15]
print('Odds:', odds)
print('Slice 3-8:', odds[3:9])
```

Output:

```
Odds: [ 1  3  5  7  9 11 13 15 17 19 21 23 25 27 29]
Slice 3-8: [ 7  9 11 13 15 17]
```

Numpy 1D-3: Create a 1D array of numbers from 10 to 100 with a step of 10. Print the last element using negative indexing.

```
import numpy as np
a = np.arange(10,101,10)
print('Array:', a)
print('Last element:', a[-1])
```

Output:

```
Array: [ 10  20  30  40  50  60  70  80  90 100]
Last element: 100
```

Numpy 1D-4: Create a 1D array of numbers 1 to 12. Reshape it into a (3,4) array.

```
import numpy as np
a = np.arange(1,13)
print('1D:', a)
print('Reshaped (3,4):\n', a.reshape(3,4))
```

Output:

```
1D: [ 1  2  3  4  5  6  7  8  9 10 11 12]
Reshaped (3,4):
 [[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
```

Numpy 1D-5: Create an array [5,10,15,20,25]. Broadcast it by adding 5 to all elements.

```
import numpy as np
a = np.array([5,10,15,20,25])
print('Before:', a)
print('After add 5:', a + 5)
```

Output:

```
Before: [ 5 10 15 20 25]
After add 5: [10 15 20 25 30]
```

Numpy 1D-6: Create a 1D array of 12 elements and reshape it into (2,6).

```
import numpy as np
a = np.arange(1,13)
print('Reshaped (2,6):\n', a.reshape(2,6))
```

Output:

```
Reshaped (2,6):
 [[ 1  2  3  4  5  6]
 [ 7  8  9 10 11 12]]
```

Numpy 1D-7: Create a 1D array of numbers from 50 to 60. Slice the first 5 elements.

```
import numpy as np
a = np.arange(50,61)
print('First 5:', a[:5])
```

Output:

```
First 5: [50 51 52 53 54]
```

Numpy 1D-8: Create a 1D array [2,4,6,8,10]. Broadcast it by multiplying with 3.

```
import numpy as np
a = np.array([2,4,6,8,10])
print('After *3:', a * 3)
```

Output:

```
After *3: [ 6 12 18 24 30]
```

Numpy 1D-9: Create a 1D array of numbers from 1 to 9. Reshape it into a (3,3) 2D array.

```
import numpy as np
a = np.arange(1,10)
print(a.reshape(3,3))
```

Output:

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
```

Numpy 1D-10: Create a 1D array of 20 numbers. Slice elements from index 5 till the end.

```
import numpy as np
a = np.arange(1,21)
print('Slice 5 to end:', a[5:])
```

Output:

```
Slice 5 to end: [ 6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]
```

Numpy 2D-1: Create a 2D array of shape (3,3) with numbers 1–9. Print the element at row 2, col 3.

```
import numpy as np
a = np.arange(1,10).reshape(3,3)
print(a)
print('Row2,col3 (1-based):', a[1,2])
```

Output:

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
Row2,col3 (1-based): 6
```

Numpy 2D-2: Create a 2D array of shape (4,4) with numbers 1–16. Slice the first two rows.

```
import numpy as np
a = np.arange(1,17).reshape(4,4)
print('First two rows:\n', a[:2])
```

Output:

```
First two rows:
 [[1 2 3 4]
 [5 6 7 8]]
```

Numpy 2D-3: Create a 2D array of shape (3,5) with numbers from 10 to 24. Slice the last column.

```
import numpy as np
a = np.arange(10,25).reshape(3,5)
print(a)
print('Last column:', a[:, -1])
```

Output:

```
[[10 11 12 13 14]
 [15 16 17 18 19]
 [20 21 22 23 24]]
Last column: [14 19 24]
```

Numpy 2D-4: Create a 2D array of shape (2,6). Reshape it into (3,4).

```
import numpy as np
a = np.arange(1,13).reshape(2,6)
print('Original (2,6):\n', a)
print('Reshaped (3,4):\n', a.reshape(3,4))
```

Output:

```
Original (2,6):
 [[ 1  2  3  4  5  6]
 [ 7  8  9 10 11 12]]
Reshaped (3,4):
 [[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
```

Numpy 2D-5: Create a 2D array (3×3). Slice the first row.

```
import numpy as np
a = np.arange(1,10).reshape(3,3)
print('First row:', a[0])
```

Output:

```
First row: [1 2 3]
```

Numpy 2D-6: Create a 2D array (4×4). Slice the last two rows and last two columns (bottom-right block).

```
import numpy as np
a = np.arange(1,17).reshape(4,4)
```

```
print(a)
print('Bottom-right 2x2:\n', a[-2:, -2:])
```

Output:

```
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]
 [13 14 15 16]]
Bottom-right 2x2:
 [[11 12]
 [15 16]]
```

Numpy 2D-7: Create a 2D array of shape (2,3). Broadcast by adding [10,20,30] to each row.

```
import numpy as np
a = np.arange(1,7).reshape(2,3)
print('Before:\n', a)
print('After broadcast:\n', a + np.array([10,20,30]))
```

Output:

```
Before:
 [[1 2 3]
 [4 5 6]]
After broadcast:
 [[11 22 33]
 [14 25 36]]
```

Numpy 2D-8: Create a 2D array (3×3). Print the diagonal elements using indexing.

```
import numpy as np
a = np.arange(1,10).reshape(3,3)
print('Diagonal:', np.diag(a))
```

Output:

```
Diagonal: [1 5 9]
```

Numpy 2D-9: Create a 2D array (5×2) with numbers 1–10. Reshape it into (2,5).

```
import numpy as np
a = np.arange(1,11).reshape(5,2)
print('Original (5,2):\n', a)
print('Reshaped (2,5):\n', a.reshape(2,5))
```

Output:

```
Original (5,2):
 [[ 1  2]
 [ 3  4]
 [ 5  6]
 [ 7  8]
 [ 9 10]]
Reshaped (2,5):
 [[ 1  2  3  4  5]
 [ 6  7  8  9 10]]
```

Numpy 2D-10: Create a 2D array (3×4). Slice all rows but only columns 1 and 2.

```
import numpy as np
a = np.arange(1,13).reshape(3,4)
```

```
print(a)
print('Cols 1 and 2:\n', a[:,1:3])
```

Output:

```
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
Cols 1 and 2:
 [[ 2  3]
 [ 6  7]
 [10 11]]
```

Numpy 3D-1: Create a 3D array of shape (2,3,4) with numbers 1–24. Print element at [1,2,3].

```
import numpy as np
a = np.arange(1,25).reshape(2,3,4)
print('Element [1,2,3]:', a[1,2,3])
```

Output:

```
Element [1,2,3]: 24
```

Numpy 3D-2: Create a 3D array of shape (2,2,3). Slice the first block (index 0 along axis 0).

```
import numpy as np
a = np.arange(1,13).reshape(2,2,3)
print('First block:\n', a[0])
```

Output:

```
First block:
 [[1 2 3]
 [4 5 6]]
```

Numpy 3D-3: Create a 3D array (3,3,3) with numbers 1–27. Reshape it into (9,3).

```
import numpy as np
a = np.arange(1,28).reshape(3,3,3)
print('Reshaped (9,3):\n', a.reshape(9,3))
```

Output:

```
Reshaped (9,3):
 [[ 1  2  3]
 [ 4  5  6]
 [ 7  8  9]
 [10 11 12]
 [13 14 15]
 [16 17 18]
 [19 20 21]
 [22 23 24]
 [25 26 27]]
```

Numpy 3D-4: Create a 3D array (2,3,3). Slice the second row of the first block.

```
import numpy as np
a = np.arange(1,19).reshape(2,3,3)
print('Second row of first block:', a[0,1])
```

Output:

```
Second row of first block: [4 5 6]
```

Numpy 3D-5: Create a 3D array (2,3,3). Broadcast by adding [5,10,15] to each row.

```
import numpy as np
a = np.arange(1,19).reshape(2,3,3)
print('Before:\n', a[0])
print('After row-add:\n', a[0] + np.array([5,10,15]))
```

Output:

```
Before:
 [[1 2 3]
 [4 5 6]
 [7 8 9]]
After row-add:
 [[ 6 12 18]
 [ 9 15 21]
 [12 18 24]]
```

Numpy 3D-6: Create a 3D array (2,2,4) with numbers from 1 to 16. Slice the last two columns of all blocks.

```
import numpy as np
a = np.arange(1,17).reshape(2,2,4)
print('Last two columns of all blocks:\n', a[:,:,-2:])
```

Output:

```
Last two columns of all blocks:
 [[[ 3  4]
  [ 7  8]]

 [[11 12]
  [15 16]]]
```

Numpy 3D-7: Create a 3D array (2,3,2). Print the element at [0,1,1].

```
import numpy as np
a = np.arange(1,13).reshape(2,3,2)
print('Element [0,1,1]:', a[0,1,1])
```

Output:

```
Element [0,1,1]: 4
```

Numpy 3D-8: Create a 3D array (3,2,2). Reshape it into (2,3,2).

```
import numpy as np
a = np.arange(1,13).reshape(3,2,2)
print('Reshaped to (2,3,2):\n', a.reshape(2,3,2))
```

Output:

```
Reshaped to (2,3,2):
 [[[ 1  2]
  [ 3  4]
  [ 5  6]]

 [[ 7  8]
  [ 9 10]
  [11 12]]]
```

Numpy 3D-9: Create a 3D array (2,3,3). Slice the last row of every block.

```
import numpy as np
a = np.arange(1,19).reshape(2,3,3)
print('Last row of every block:\n', a[:, -1, :])
```

Output:

```
Last row of every block:
 [[ 7  8  9]
 [16 17 18]]
```

Numpy 3D-10: Create a 3D array (2,2,3). Broadcast by multiplying with 10.

```
import numpy as np
a = np.arange(1,13).reshape(2,2,3)
print('After *10:\n', a * 10)
```

Output:

```
After *10:
 [[[ 10  20  30]
  [ 40  50  60]]

 [[ 70  80  90]
  [100 110 120]]]
```